

Md. Rifat Hossain

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EDUCATION

United International University <i>Bachelor of Science in Computer Science and Engineering</i> — CGPA: 3.87/4.00(German Grade: ≈ 1.1) July 2022 – Present	Dhaka, Bangladesh
• Completed 125 out of 137 credits.	
Mirpur Cantonment School and College <i>Higher Secondary Certificate Examination</i> — GPA: 5.00/5.00(German Grade: ≈ 1.0)	Dhaka, Bangladesh
Adamjee Cantonment Public School <i>Secondary School Certificate Examination</i> — GPA: 4.89/5.00(German Grade: ≈ 1.1)	Dhaka, Bangladesh

EXPERIENCE

Undergraduate Teaching Assistant <i>United International University</i>	Mar. 2025 – Present <i>Dhaka, Bangladesh</i>
- Selected as an Undergraduate Teaching Assistant for three separate appointments: Mar. 2025–Jul. 2025, Aug. 2025–Oct. 2025, and Jul. 2026–Present. - Supported faculty members in conducting laboratory classes by assisting students with programming concepts, practical tasks, and course-related technical problems. - Provided individualized academic guidance and counseling, reviewed laboratory topics, and conducted supplementary tutorial sessions for students requiring additional support.	

PROJECT EXPERIENCE

LifeLens AI <i>React, FastAPI, MySQL, LangChain, AI/LLM</i> <i>Full-Stack & AI Developer</i> <i>Team Product</i> GitHub	
- Developed an AI-powered personal decision assistant covering meal planning, productivity, career and learning guidance, and financial planning. - Implemented personalized AI recommendations and conversational assistance through a React frontend, FastAPI backend, and MySQL database.	
Sentinel Intelligence <i>YOLO, OpenCV, Python, Flask, ESP32, MySQL, IoT</i> <i>AI & IoT Developer</i> <i>Solo Product</i> GitHub	
- Built an AI-powered smart safety and emergency response system for warehouses and industrial environments, integrating computer vision, IoT, embedded systems, and web technologies. - Implemented automated fire detection, tracking, and suppression alongside gas, temperature, humidity, and seismic monitoring with automated emergency responses. - Developed a web-based monitoring and control platform with real-time safety status, device control, alerts, historical logging, and data visualization.	
People Flow Detection & Tracking <i>YOLOv8s, ByteTrack, OpenCV, Python, Supervision</i> <i>Computer Vision Developer</i> <i>Solo Product</i> GitHub	
- Developed a computer vision system for detecting and tracking people in video while counting directional entry and exit events. - Implemented YOLOv8s with ByteTrack for persistent multi-object tracking and trajectory-based line-crossing analysis. - Generated movement heatmaps and annotated videos with tracking IDs and real-time IN/OUT counters using OpenCV.	
Football Game <i>Java, JavaFX, Object-Oriented Programming</i> <i>Software Developer</i> <i>Team Product</i> GitHub	
- Developed a 2D multiplayer football game with competitive 1v1 matches, player management, ranking, and reward systems. - Implemented player trading and upgrades, in-game messaging, a player marketplace, and interactive reward mechanics using JavaFX and object-oriented design.	

TECHNICAL SKILLS

Core Skills: Software Development, Full-Stack Development, REST API Development, Database Systems, Object-Oriented Programming, Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Computer Vision, Data Analysis, Retrieval-Augmented Generation (RAG), Agentic AI, IoT, Embedded Systems, Web Scraping
Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS, Dart
Web Frameworks: React.js, Next.js, FastAPI, Node.js, Express.js, NestJS, Tailwind CSS
AI & ML: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO, Ultralytics, OpenCV, LangChain, LangGraph, OpenAI, Hugging Face, spaCy, MLflow, NumPy, Pandas, Matplotlib, Seaborn
Database & ORM: MySQL, PostgreSQL, SQLAlchemy
Developer Tools: Git, GitHub, Docker, Jupyter Notebook, Google Colab, Google Cloud Platform, Postman, Swagger/OpenAPI
Others: Flutter, JavaFX, Firebase, Clerk, Better-Auth

HONOURS & AWARDS

UIU CSE Project Show Jan. 2025
1st Runner Up – Advanced Object-Oriented Programming United International University
UIU Merit-Based Scholarship
United International University

- 100% Merit Scholarship Holder – 5 times (4th, 5th, 6th, 10th & 12th Trimesters)
- 50% Merit Scholarship Holder – 2 times (7th & 9th Trimesters)

School Athletics 2018
1st Prize – 100m & 200m Sprint College Level

VOLUNTEERING

Volunteer Apr. 2023 – Present
United Nations Youth and Students Association of Bangladesh (UNYSAB) Bangladesh

- Contribute to youth-led community initiatives, social awareness programs, and humanitarian activities organized by UNYSAB.

Relief for Flood-Affected People Aug. 2024
United Nations Youth and Students Association of Bangladesh (UNYSAB) Feni, Bangladesh

- Participated in a relief mission supporting communities affected by severe flooding in Feni.

Eid for Street Children Mar. 2024
United Nations Youth and Students Association of Bangladesh (UNYSAB) Dhaka, Bangladesh

- Participated in a two-day community initiative providing iftar meals and support to underprivileged street children.

LANGUAGES

Bangla: Mother Tongue
English: Fluent
Hindi: Speaking & Listening